

Indian Institute of Technology, Bhilai



CSL351: Computer Networks

Assignment - 9 (Network Layer - Routing)

Full Marks: 30

Deadline: 30/03/2026, Time: 11:59 PM

Submission Instructions:

- 1) Answer all the questions.
- 2) Deliverables in a .zip:
 - a) Readable Report [**5 Marks for report quality**] enumerating steps followed with screenshots for each of the important steps.
 - i) Put the screenshots in the report for better clarity
 - ii) Explain everything that has been asked in your report and highlight the important parts, and follow the instructions given in the question carefully.
 - b) The .pkt file of the CISCO packet tracer.
- 3) You can use LaTeX/Word to make the PDF.
- 4) The naming of the zip file should strictly follow the given format:
<Roll_No>_<Name>_<Assignment No>. If your name is Alex, your roll number is B23CS055, then the filename should be: B23CS055_Alex_09.zip
- 5) **Any plagiarism case will be considered an unethical practice, and appropriate action will be taken against them.**

Objectives:

1. Understand Network Topology Design
2. Implement LAN and WAN Integration
3. Hands-on Experience with Cisco Packet Tracer
4. IP Addressing and Configuration
5. Router Configuration Skills
6. Network Testing and Troubleshooting

Design and simulate a network in **Cisco Packet Tracer** where:

[25 Marks]

- 3 routers (R1, R2, R3) are connected in a triangle (full mesh)

- 2 of the router is connected to switches (S1, S2).
- Switches connect 5 end devices (PCs/users) (PC1, PC2, PC3, PC4, PC5) and the 6th end device, PC6, is connected to a router as shown below.

Correct Network Structure:

1. Router Connections (WAN):
 - a. R1 <-> R2
 - b. R2 <-> R3
 - c. R3 <-> R1
2. LAN Connections:
 - a. R1 -> S1 -> PC1, PC2, PC3
 - b. R3 -> S2 -> PC4, PC5
 - c. R2 -> PC6

Part A: Topology Design

[10 Marks]

1. Place:
 - 3 Routers (e.g., 2911)
 - 2 Switches (e.g., 2960)
 - 6 PCs
2. Connect:
 - Router-to-Router Link: **Serial or Gigabit links**
 - Router-to-Switch Link: **Copper straight-through**
 - Switch-to-PC Link: **Copper straight-through**
 - Router-to-PC Link: **Gigabit link**

Part B: Configuration

[10 Marks]

1. **Configure PCs:**
 - a. Assign IP + Subnet Mask
 - b. Set Default Gateway (Router IP)
2. **Configure Routers:**
 - a. Assign IP to all interfaces
 - b. Enable interfaces (no shutdown)
 - c. Static Routing

Part C: Verification

[5 Marks]

1. Ping:
 - a. PC1 -> PC3
 - b. PC2 -> PC5
 - c. PC6 -> PC4
2. Use simulation mode to check packet flow.

Note: For each configuration and verification, you must explain the process and attach the corresponding screenshots to the report.